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1 function [echogramstruct,fEcho,RangeBinned,SampleBinned] = ✓
calcEchogram90kHzBinned(Config, Data, WaterProp, Options)
2 %function [echogram90bin1,echogram90bin2,echogram90bin3, ✓
echogram90bin4,echogram90bin5] = calcEchogram90kHzBinned(Config, Data, ✓
NBINF)
3 %function [echogram90bin1,echogram90bin2,echogram90bin3, ✓
echogram90bin4,echogram90bin5] = calcEchogram90kHzBinned(Config, Data)
4 %
5 % S. Nylund, 201908
6 % G. Cutter, 201909
7
8 %% OPTIONS
9 % Struct Options should include values for:
10 % Options.retRawData % DEFAULT=0
11 % Options.NBINF %default=5
12
13 %retRawData = 1;
14 %if(isempty(Options.retRawData))
15 if(isempty(Options))
16     retRawData = 0; % retRawData = 1;
17     NBINF      = [];
18 else
19     retRawData = Options.retRawData;
20     NBINF      = Options.NBINF;
21 end
22
23 %% INIT
24 echogramstruct = []; % in case we want more than default 5 freq-bins
25 % echogram90bin1 = [];
26 % echogram90bin2 = [];
27 % echogram90bin3 = [];
28 % echogram90bin4 = [];
29 % echogram90bin5 = [];
30 try
31     soundVel = WaterProp.soundVel;
32 catch
33     soundVel = 1500;
34 end
35
36 BD=Config.echo_blanking;
37
```

```
38 if Config.echo_frequency1 == 90 && Config.echo_pulseComp1 == 1
39     nEcho=1;
40 else
41     if isfield(Config,'echo_frequency2') && Config.echo_frequency2 ✓
42         == 90 && Config.echo_pulseComp2 == 1
43             nEcho=2;
44         else
45             if isfield(Config,'echo_frequency3') && Config. ✓
46                 echo_frequency3 == 90 && Config.echo_pulseComp3 == 1
47                     nEcho=3;
48                 else
49                     disp('No 90 kHz pulse compressed data');
50                     return;
51                 end
52             end
53         end
54     end
55 eval(['sampleRate = Config.echo_rawSampleRate' num2str(nEcho) ';' ])
56 eval(['nRawSamp    = Config.echo_nRawSamp'      num2str(nEcho) ';' ])
57 eval(['DataI = Data.RawEcho' num2str(nEcho) '_90kHz_DataI;'])
58 eval(['DataQ = Data.RawEcho' num2str(nEcho) '_90kHz_DataQ;'])
59 eval(['Range = Data.RawEcho' num2str(nEcho) '_90kHz_Range;'])
60 eval(['Zxmit = Data.RawEcho' num2str(nEcho) '_90kHzTx_DataI + ✓
61 j*Data.RawEcho' num2str(nEcho) '_90kHzTx_DataQ;']);
62 Zxmit0 = Zxmit;
63 Zxmit=[Zxmit zeros(1,100)]; % Why the extra zeros?
64 %%
65 if nRawSamp > 0,
66     sampDist90 = soundVel/(2*sampleRate);
67     Nsamp = size(DataI,1);
68 else
69     disp('No raw echo data');
70     return;
71 end
72
73
74 %% DEBUGGING
75 fprintf('sizes\n');
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76 fprintf('DataI    %d %d\n', size( DataI ) );
77 fprintf('DataQ    %d %d\n', size( DataQ ) );
78 fprintf('Zxmit0    %d %d\n', size( Zxmit0 ) );
79 fprintf('Zxmit     %d %d\n', size( Zxmit ) );
80 fprintf('Range     %d %d\n', size( Range ) );
81
82
83 %% range
84 N=length(Data.RawEcho1_90kHz_SampleRate);
85 Z=DataI+j*DataQ;
86 pcLin=zeros(N,length(real(Z(1,:)))-length(Zxmit0));
87 %r=0:(length(pcLin(1,:))-1);
88 %r=0:(length(pcLin(1,:)));
89 r=0:(length(pcLin(1,:))-2);
90 r=r*(WaterProp.soundVel*0.5)/Config.echo_rawSampleRate1;
91
92
93
94 %% Calculate complex bandpass filters
95 load LPfilter % Load low pass filter
96 ECHO_REMOVE_ENDS = 1.875;
97 recvBW = 0.5; % 50% bandwidth
98 %% Number of frequency bins
99 if isempty(NBINF)
100     nbinf = Config.echo_nbinf1;
101 else
102     nbinf = NBINF;
103 end
104
105 %%
106 fFc = 1.0e6/11;
107 ECHO_FILTER_PC_LENGTH = length(B); % B is from LPfilter
108 coeffComplexBinnedFrequency = zeros(nbinf,ECHO_FILTER_PC_LENGTH);
109
110 L2 = size(Data.RawEcho1_90kHzTx_DataI,2);
111 L2filtered = round(L2/nbinf + ECHO_FILTER_PC_LENGTH);
112 nShiftBinPos = floor((0:nbinf-1)*L2/nbinf);
113
114 if Config.fwVersionDoppler < 2212
115     fAdjPulseCompAmp = 10.0*log10(L2) - 30.0; % Adjustment ✓
116     (erroneous) used in firmware [dB]
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116 else
117     fAdjPulseCompAmp = 10.0*log10(L2filtered) - 30.0; % Adjustment✓
used in firmware [dB]
118 end
119
120 for k=0:nbinf-1
121     fF0 = (ECHO_REMOVE_ENDS/nbinf)*(k-floor(nbinf/2.0));
122     F0(k+1) = fF0;
123     fEcho(k+1) = fFc*(1+fF0*recvBW/2.0);
124     coeffComplexBinnedFrequency(k+1,:) = B.*exp(j*fF0*pi*(0:✓
ECHO_FILTER_PC_LENGTH-1));
125
126     filtXmit(k+1,:) = filter(coeffComplexBinnedFrequency(k+1,:), 1, ✓
Zxmit);
127
128     nCenter = L2/2+ECHO_FILTER_PC_LENGTH/2 + L2*((floor(nbinf/2.0)-✓
k)/nbinf);
129     startIndex = max(0,round(nCenter-L2filtered/2));
130     endIndex = min(size(filtXmit,2)-1,round(nCenter+L2filtered/2));
131     len = 1+endIndex-startIndex;
132
133     if Config.fwVersionDoppler < 2212
134         % Clip/saturate filter coefficients in order to replicate✓
firmware bug
135         preFiltXmit(k+1,1:len) = complexSaturate(filtXmit(k+1,1+✓
(startIndex:endIndex)), 1.0, -1.0);
136     else
137         preFiltXmit(k+1,1:len) = filtXmit(k+1,1+(startIndex:✓
endIndex));
138     end
139 end
140
141 fNSampPreStart = ((sampleRate * Config.echo_transmitLength1 * 1.0e-✓
3/nbinf) + ECHO_FILTER_PC_LENGTH)/2.0;
142 indStartSamp = floor(0.5 + (sampleRate * BD/(soundVel/2)) -✓
fNSampPreStart);
143 startInd90 = indStartSamp;
144 FiltDelay = len;
145
146 % Calculate the number of samples per bin
147 samplesPerCell = Config.echo_binSize/sampDist90;
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148 BinLength=zeros(1,Config.echo_numBins);
149 BinLength(1) = round(samplesPerCell);
150 for i=2:Config.echo_numBins
151     BinLength(i) = round((i*Config.echo_binSize - sum(BinLength(1:i-1))
152     ))*sampDist90)/sampDist90);
153 end
154 echogram90raw = [];
155
156 % % Range = Range(Range>=0);
157 RangeBinned = zeros(1, Config.echo_numBins);
158 SampleBinned = RangeBinned;
159
160 for i=1:nbinf
161
162     %% Creates an echogram array for each frequency bin
163     eid = sprintf('%2.2d',i); % echogram bin identifier
164     eval(['echogram90bin' eid ' = [];']); %init empty echogram
165     eval(['echogram90raw' eid ' = [];']); %init empty echogram
166     echogrambinname = ['echogram90bin' eid];
167
168     % Apply complex filter which performs filtering and convolution
169     %X = DataI(: , nShiftBinPos(i)+(startInd90:(end-nShiftBinPos
170     (end))) )' +j*DataQ(:,nShiftBinPos(i)+(startInd90:(end-nShiftBinPos
171     (end))))';
172     INDX2 = nShiftBinPos(i)+(startInd90:(size(DataI,2)-
173     nShiftBinPos(size(nShiftBinPos,2))));
174     X = DataI(: , INDX2 )' +j*DataQ(:,nShiftBinPos(i)+(startInd90:
175     (end-nShiftBinPos(end))))';
176     DataFilt = filter(preFiltXmit(i,:), 1, X)';
177
178     %% Range init
179     % NO. MOVE THIS -- % RangeBinned = Range( :, INDX2);
180     %no RangeBinned = Range( :, INDX2);
181     %no RangeBinned = r;
182     %RangeBinned = Range;
183     %RangeBinned = Range( :, INDX2); % or (Range>=0); %?
184     %Range = Range( :, INDX2); % or (Range>=0); %?
185     Range = Range(Range>=0);
186     SampleBinned = 1:length(Range);
187

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184     if retRawData,
185         echogram90bin = zeros(Nsamp, Config.echo_numBins);
186         echogram90raw = [echogram90raw; 10*log10(DataFilt.*conj(
(DataFilt)))]';
187
188         %% Range for raw echogram
189         % %      %no RangeBinned = Range( :, INDX2);
190         % %      %no RangeBinned = r;
191         % %      %RangeBinned = Range;
192         % %      RangeBinned = Range(Range>=0);
193         % %      SampleBinned = 1:length(Range);
194         RangeBinned = Range;
195         if( length(RangeBinned) > size(DataFilt,2) )
196             RangeBinned = RangeBinned( 1 : size(DataFilt,2) );
197         end
198     end
199
200     if retRawData,
201         echogram90raw = echogram90raw';
202         %eval(['echogram90bin' num2str(i) ' = echogram90raw;']);
203         eval(['echogram90bin' eid ' = echogram90raw;']);
204         echogram90raw = [];
205     else
206         for n=1:Nsamp
207             k = FiltDelay; % Sample number for start of first
echogram bin. Blanking has already been accounted for.
208
209             for m=1:Config.echo_numBins
210                 samplesInCell = k:(k+BinLength(m)-1);
211
212                 Z = DataFilt(n,samplesInCell);
213                 sumZZ = sum(Z.*conj(Z));
214                 if sumZZ == 0,
215                     % Get rid of log10 of zero messages
216                     sumZZ = 1e-16;
217                 end
218                 echogram90bin(n,m) = fAdjPulseCompAmp + 10*log10(
(sumZZ/BinLength(m))); % Average. Using 10*log10 since Z is squared
219
220
221                 %% Range for binned echogram

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```
222         % %RangeBinned(m) = mean(Range(samplesInCell));
223         % %SampleBinned(m) = mean(samplesInCell);
224         if Config.fwVersionDoppler < 2212
225             %no RangeBinned(m) = mean(Range(samplesInCell + L2));
226             %RangeBinned(m) = Config.echo_binSize + mean(Range(startInd90 + samplesInCell + FiltDelay));
227             %RangeBinned(m) = mean(Range(startInd90 + samplesInCell + FiltDelay));
228             RangeBinned(m) = mean(Range( samplesInCell ));
229             SampleBinned(m) = mean(samplesInCell) + L2;
230         else
231             %no RangeBinned(m) = mean(Range(samplesInCell + L2filtered));
232             %RangeBinned(m) = Config.echo_binSize + mean(Range(startInd90 + samplesInCell + FiltDelay));
233             %RangeBinned(m) = mean(Range(startInd90 + samplesInCell + FiltDelay));
234             RangeBinned(m) = mean(Range( samplesInCell ));
235             SampleBinned(m) = mean(samplesInCell) + L2filtered;
236         end
237         fprintf('samplesInCell (%d : %d), Range %.2f\n', samplesInCell(1), samplesInCell(end), RangeBinned(m));
238
239         k = k+BinLength(m);
240     end
241 end
242 eval(['echogram90bin' eid ' = echogram90bin;']);
243
244 end
245
246 echogramstruct.(echogrambinname) = eval(['echogram90bin' eid]);
247
248 %% DEBUGGING
249 [m_echo n_echo] = size( echogramstruct.(echogrambinname) );
250 fprintf('\nsizes, end of loop.\n');
251 fprintf('DataI      %d %d\n', size( DataI ) );
252 fprintf('echog      %d %d\n', m_echo, n_echo );
253 fprintf('Range      %d %d\n', size( Range ) );
254 fprintf('r          %d %d\n', size( r ));
```

```
255     fprintf('r-n_echo  %d\n', size( r ,2 ) - n_echo );
256     fprintf('R-n_echo  %d\n', size( r ) - n_echo );
257
258 end
259
260
261
262 fprintf('fBinnedEcho  %.1f\n', fEcho)
263
264 fprintf('\nFunction calcEchogram90kHzBinned completed.\n');
265
```